AI CRASH DETECTION SOFTWARE

Presented By,
TEAM 4

TEAM MEMBERS

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Project Overview

- Cutting-edge crash detection AI system
- Enhanced accuracy and responsiveness
- Multi-platform compatibility (PC & Mobile)
- Real-time crash detection & emergency response

SYSTEM DESIGN DETAILS

Development Tools & Technologies

- **IDE:**
  - PyCharm for PC application
  - Android Studio for mobile app development
- **Languages:**
  - Python for PC-based application
  - Dart (Flutter framework) for mobile application
  - Java for Android-specific implementations
- **Backend Infrastructure:**
  - Firebase for database management, real-time synchronization, and app hosting
  - Firebase Cloud Messaging (FCM) for notifications
- **Frontend:**
  - Tkinter or CustomTkinter for PC GUI
  - Flutter for mobile app interface

AI FRAMEWORKS & LIBRARIES

AI Model Development

- **Ultralytics YOLOv8**:
  - State-of-the-art object detection model
  - Optimized for crash detection scenarios
- **Torch & Torchvision:**
  - Core PyTorch framework used for training and deployment
- **OpenCV-python-headless:**
  - Efficient library for real-time image and video stream processing
- **Dataset:**
  - Over 8,600 labeled images and videos for robust crash event detection
  - Diverse scenarios to enhance model generalization

KEY FEATURES

System Features

1.Real-Time Crash Detection:

- •Processes video feeds to identify crash events immediately
- •High accuracy with minimal false positives and negatives

2.Automated Notifications:

- •Sends alerts to emergency responders via Firebase Cloud Messaging
- •Ensures timely action in critical situations

3.Cross-Platform Integration:

- •Synchronization between PC and mobile applications
- •Unified data and seamless monitoring experience

4.Image & Video Processing:

- •Handles high-resolution streams efficiently using OpenCV

5.Performance Analytics:

- •Tracks detection accuracy, system responsiveness, and user engagement
- •Provides insights for continuous optimization



ADVANTAGES

System Advantages

- **Efficiency:** Real-time operation minimizes delays in emergency response
- **Scalability:** Supports expanding datasets and system complexity
- **Accuracy:** Advanced AI models reduce false positives and negatives
- **Cost-Effective:** Cloud-based infrastructure reduces hardware costs
- **Data Security:** Implements robust Firebase security protocols for data privacy
- **User-Friendly:** Intuitive design ensures ease of use for both administrators and end-users

TECHNOLOGIES & TOOLS

Technologies Used

- **IDE:**
  - PyCharm for development and debugging of PC application
  - Android Studio for mobile app development
- **Frameworks:**
  - Flutter for creating cross-platform mobile apps
  - PyTorch for building and training AI models
- **Backend Infrastructure:**
  - Firebase for cloud hosting, database management, and notifications
- **Image & Video Processing:**
  - OpenCV-python-headless for efficient handling of visual data
- **AI Model Version:** YOLOv8 for crash detection

WORKFLOW DIAGRAM

1
Data Collection
Images and videos labeled.

2
Model Training
YOLOv8 trained using PyTorch.

3
Video Stream Input
Live feeds processed using OpenCV.

4
Real-time Detection
AI model identifies crash events.

5
Notification Trigger
Firebase sends alerts.

6
Result Display
Outputs shown on PC and mobile.

IMPLEMENTATION DETAILS

Development:

- Setting up IDEs (PyCharm, Android Studio) and Flutter environment
- Training AI models using YOLOv8
- Backend integration with Firebase

Testing:

- Extensive testing with diverse crash scenarios
- Performance evaluation and debugging

Deployment:

- Deployment on PC and mobile platforms
- Final validation in real-world environments

CONCLUSION

Conclusion

- Revolutionizing crash detection with AI-driven solutions
- Enhanced real-time responsiveness and multi-platform usability
- Scalable system ready for future advancements
- Aimed at improving emergency response efficiency and saving lives

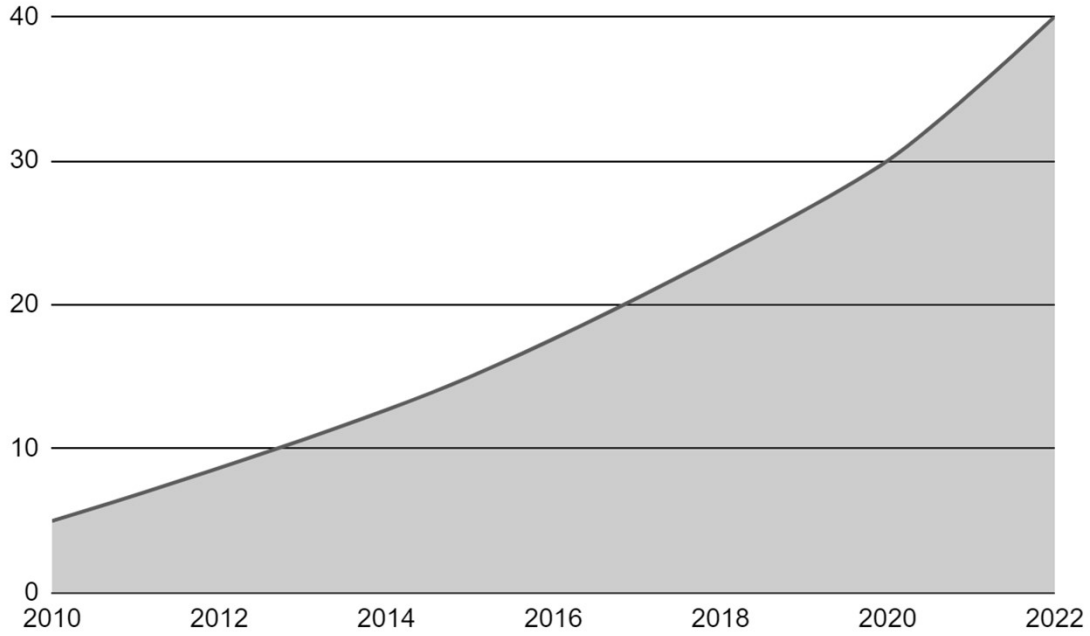
RÉSULTATS PRÉVUS

20 000

Revenu attendu pour 2022

100

Nouveaux employés pour 2022



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